

Parcel Reality Bench

What it is, what the first real run found, what got fixed, what to post, and what happens next.

CLAUDE · PILOT	RULES PIPELINE	FACE VALUE
82.9	94.5	45.7
claude.ai Default tier · 9 items · v1.1.0	200-line deterministic baseline	naive baseline

BOTTOM LINE

The first real Claude run landed 11.6 points below a simple rules pipeline on the hard tier. About 4.5 of those points were a bug in the benchmark, not in Claude. With the bug removed, Claude scores 87.4: still 7 points behind the pipeline, on misses that are real.

The bug is fixed (generator v1.1.1). The benchmark caught a flaw in itself on its first live run, which is the strongest evidence yet that the scoring and failure analysis work.

What it is

A benchmark that tests whether an AI can value a house from the kind of records that actually exist: county exports, state sales-disclosure files, MLS feeds, and assessor websites, as received, with their errors. Territory: St. Joseph, LaPorte, Elkhart (Indiana) and Berrien, Cass (Michigan). The state line runs through the middle of the market, which is where most valuation errors start.

Every item is a subject property and six comparable sales. The model must return a value, an 80% range, which comps it threw out, which traps it found, and which state and county the property is in.

TRAP	WHAT HAPPENS IN REAL RECORDS	HARD-TIER VERSION (NO HINTS)
Price scale	Indiana sales-disclosure files store price in an implied-decimal layout	Raw field only; the model must know the layout
Related-party sale	Family transfers at ~50% of market	Normal warranty deed; only a shared surname and the low price give it away
Multi-parcel sale	One price covering several parcels	Three rows, each showing the full price, as in real state files
Keying error	Extra zero typed into a price	A single 10x error
Duplicate record	Same sale from two sources	Different date format and an undashed parcel number
Cross-border comp	An Indiana comp used for a Michigan house	Same
Jurisdiction inversion	Owner mails from one state, house sits in the other	Blank situs county, no glossary

Score = 30% valuation accuracy + 20% range calibration + 25% correct comp exclusion + 15% correct trap flags + 10% correct jurisdiction.

What is built

PIECE	WHAT IT DOES	STATUS
Generator	Makes deterministic test items across 7 trap types and 2 difficulty tiers	v1.1.1, verified
Scorer	Composite score plus a per-trap failure map	Python and in-browser versions agree to 6 decimals
Baselines	Oracle, rules pipeline, face-value	Run on all four splits
Model adapters	Anthropic, OpenAI, Gemini; no extra software	Built; needs your keys
Receipt ledger	Hash chain over every prompt, answer, and score	1,574 receipts, 0 mismatches
Sealed holdout	200 hidden items per tier, sealed before any model runs	Committed
Leaderboard	Three-page site on Neuruh v1 design law	Built; deploys to bench.neuruh.com
Auto loop	Finds new models, runs, verifies, publishes, drafts posts	Tested end to end
Live runner	Browser page that runs Claude with no setup	Live on claude.ai

What is actually cool

- It audited itself. The first real run exposed a flaw in the test data within minutes. The failure map pointed straight at it. Benchmarks that cannot find their own bugs publish them.
- Nobody can quietly change the numbers. Every answer is fingerprinted and chained. Your downloaded result file was re-verified in Python: chain intact, every score reproduced exactly.
- Nobody can train against it. The holdout answers are sealed with a hash published before any model runs. Revealing them later proves the goalposts never moved.
- Fully reproducible. Regenerating v1.1.0 reproduced the exact dataset fingerprint the pilot ran on (a51b4158...). Anyone can rebuild it and check.
- It runs itself. A new frontier model gets benchmarked the morning after release, with a post draft waiting. Nothing posts without you.
- It is ours. The traps come from 25 years of reading these records by hand. A lab cannot build this from a web scrape.

Results — pilot run, 2026-09-22

Claude on claude.ai, Default tier, 9 items of the hard public split, generator v1.1.0. Run id Live-20260922T113837-default. Receipt chain verified; ledger head c951d416...

COMPONENT	CLAUDE	WHAT IT MEASURES
Valuation accuracy	87.5	How close the value is to truth (full credit within 0%, zero at 25% off)
Range calibration	72.8	How tight the 80% range is while still containing truth
Comp exclusion F1	83.0	Threw out the bad comps and kept the good ones
Trap flag F1	82.9	Named the right traps
Jurisdiction	88.9	Right state and county
Median value error	2.9%	Typical miss on price
Range coverage	100%	Truth was inside the range every time
Composite	82.9	Weighted total

Where Claude failed

ITEM	WHAT HAPPENED	WHOSE FAULT
PRX-0002	Put a Berrien County, Michigan property in LaPorte County, Indiana, and kept a cross-border comp	Claude — real miss
PRX-0005	Threw out a normal market sale because buyer and seller were both named ROBERT	Benchmark bug
PRX-0006	Threw out a normal market sale because buyer and seller were both named EMILY	Benchmark bug
All items	Ranges wide enough to be always right, too wide to be useful	Claude — real pattern

Caveat: 9 items is a pilot, not a result. It is enough to find bugs and patterns, not enough to publish a score. Public numbers need the full split on pinned model IDs.

The fix — generator v1.1.1

Bug. Clean sales drew buyer and seller first names independently from a 10-name pool, so about 1 in 11 clean comps had a coincidental matching first name. On the hard tier, the related-party signal is a shared name plus a low price. A careful reader could fairly treat ROBERT → ROBERT as a flag. The benchmark then punished that caution.

SPLIT	CLEAN COMPS	COINCIDENTAL FIRST-NAME MATCHES (V1.1.0)	AFTER V1.1.1
public_hard	329	32	0
public	327	30	0
holdout_hard	1,103	92	0
holdout	1,106	97	0

Fix. Clean sales now share no name token between buyer and seller. Related-party traps are unchanged. All splits regenerated, holdouts re-sealed, baselines re-run, full chain re-verified: 1,574 receipts, 1,560 answers re-scored, 0 mismatches. The live runner now serves v1.1.1.

Measured cost of the bug. Re-scoring Claude’s pilot with only the two bug-induced exclusions removed: 82.9 → 87.4. The pilot file is kept in pilot/ as the record of how the bug was found.

Posts

POST A — SAFE TO SHARE NOW (X / LINKEDIN)

Built a benchmark for AI on real county property records. The Indiana–Michigan border kind: prices stored in state file formats, family sales dressed as market sales, houses whose owner mails from the wrong state.

Pilot run on Claude, 9 items: 82.9. A 200-line rules pipeline: 94.5.

Best part: two of Claude's worst misses were my fault. My generator gave unrelated buyers and sellers the same first name, and Claude read ROBERT → ROBERT as a family sale. Fair read. Fixed, v1.1.1 shipped.

Every answer hash-chained. Holdout sealed. Frontier runs next.

POST B — REPLY TO LOGAN KILPATRICK (ONLY AFTER PINNED HARD-TIER SCORES EXIST)

took you up on it. Parcel Reality Bench: county-records valuation on the IN/MI line, 7 planted data-trap classes, hash-chained receipts, sealed holdout. a 200-line rules pipeline scores 94.5 on the hard tier. best frontier model: [SCORE] ([MODEL]). top miss: [TRAP] at [RATE]%. [LINK]

POST C — LAUNCH (AFTER BENCH.NEURUH.COM IS LIVE)

Most AI benchmarks test what models know. Parcel Reality Bench tests what they notice. [N] frontier models, county records as they actually arrive, seven trap types appraisers hit weekly. Leaderboard, receipts, and failure maps: bench.neuruh.com. New models are tested the morning after release.

Rule for all three: never post a number that is not in the ledger. Post B and C stay unposted until the leaderboard shows the numbers they quote.

Next steps

#	STEP	WHO	TIME
1	Re-run the live page on v1.1.1 at 30 items; send the result file	You	20 min
2	Post A	You	5 min
3	Path 2 on the Mac: keys in Keychain, first pinned runs	You	30 min
4	Path 3: GitHub + Cloudflare via Claude Code	You + Claude Code	30 min
5	CRUCIBLE on the full HaluBench set (plan below)	Claude + Claude Code	1 session
6	Harden any trap the saturation report flags	Claude	1 session
7	Live track: real Indiana sales through DATUM, refreshed monthly	Claude Code on M1	2–3 sessions
8	Posts B and C	You	When numbers exist

CRUCIBLE → HaluBench plan

- CRUCIBLE already has a harness (`bench/run_bench.py`) that speaks HaluBench format. Its existing numbers (F1 0.923 heuristic, 0.700 Bonsai) come from a 14-item seed. That is a smoke test.
- Pull the public HaluBench set (Patronus AI), convert to the harness format, run against the live service in both modes, report precision, recall, F1, balanced accuracy, and p50/p95 latency per source subset.
- Compare against published detector results. Wrap every run in the same receipt chain as `prbench` so both benchmarks share one provenance story.

Where the money is

- Private evaluations: proptech and AVM vendors pay to run their model on the sealed holdout without exposing it.
- Certification: a “PRB-verified” mark for tools that clear a threshold, re-tested each version.
- Trap generator licensing: county-data and valuation vendors license it as a QA suite for their own pipelines. White-label from day one: branding lives in config.
- DATUM positioning: once the live track runs, DATUM sits on the same board as the frontier labs on real Michiana sales.
- Cross-vertical: the same pattern (planted traps + receipts + sealed holdout) ports to resale comps — sold vs listed, bundle lots, relists, wrong-condition comps. Same engine, new trap set.